

Play Store App Performance & Sentiment Dashboard

Power BI DataViz Challenge — Project Documentation

1. Project Overview

This dashboard was built as part of the DataViz Challenge, translating findings from my earlier Python/Pandas Exploratory Data Analysis (EDA) of the Google Play Store dataset into an interactive Power BI dashboard aimed at a business audience.

Tool used: Microsoft Power BI Desktop.

2. Business Objective

How can app publishers and product teams use category performance, pricing strategy, and user sentiment data to identify which app categories offer the best growth and monetization opportunities?

3. Dataset Details

- Source: Google Play Store Apps dataset (Kaggle) — the same dataset used in my Python EDA project.
- Apps table: 9,637 apps after cleaning (category, rating, installs, price, size, type, content rating, last updated).
- Reviews table: approximately 64,000 user reviews, each with a Sentiment label (Positive / Negative / Neutral), Sentiment Polarity, and Sentiment Subjectivity.
- Cleaning steps performed in Python (Pandas): removed a corrupted row with shifted column values, converted Price and Installs from text to numeric, handled nulls, removed exact duplicate rows.
- Additional cleaning performed in Power Query (Power BI): converted remaining text-formatted numeric fields (Price, Installs, Reviews, Rating, Last Updated), removed rows with unresolvable formatting errors (for example, abbreviated review counts such as "3.0M"), trimmed whitespace, and removed duplicate App entries (kept the row with the highest Reviews count per app, since duplicates represented the same app captured at different scrape times).
- Apps and Reviews were combined using a Left Outer merge in Power Query, joined on the App name.

4. Data Modeling Note — Fan-Out Correction

Because each app can have many reviews, merging the Apps and Reviews tables produced multiple rows per app (one per review). Early versions of the Total Installs measure and the Treemap's app count used simple SUM and COUNT, which multiplied each app's values by its number of review rows — for example, showing 3 trillion total installs and a single category with more apps than existed in the entire dataset.

This was corrected by rewriting the affected measures to calculate once per distinct app, for example:

```
Total Apps = DISTINCTCOUNT(playstore_apps_cleaned[App])
Total Installs = SUMX(VALUES(playstore_apps_cleaned[App]),
CALCULATE(MAX(playstore_apps_cleaned[Installs])))
```

After this fix, Total Installs correctly shows approximately 75 billion, and every category's app count and install total is consistent with the overall KPI cards. Average-based measures (Avg Rating, Avg Price) were unaffected, since averaging duplicate identical values does not change the result.

5. Dashboard Design

The dashboard is a single page containing 7 visuals and 3 global interactive slicers (Category, Type, Content Rating), following a KPI-first, top-to-bottom layout: title, slicers, KPI cards, then supporting charts grouped by theme (market/performance, then sentiment).

6. Visual-by-Visual Insights

1. KPI Cards — Total Apps, Total Installs, Avg Rating

Chart type: Cards

Why this visual: Gives an immediate, at-a-glance orientation to the dataset's scale before diving into detail.

Key insight: 9,637 apps in the dataset, averaging a 4.28 rating, with roughly 75 billion combined installs.

Business relevance: Sets the baseline scale for any category-level comparison made elsewhere on the dashboard.

2. App Distribution by Category (Treemap)

Chart type: Treemap

Why this visual: Treemaps communicate relative volume across many categories in one glance, using area instead of a long list of bars.

Key insight: Games, Family, and Tools are the categories with the highest number of listed apps, reflecting where developer competition is most concentrated.

Business relevance: Helps a publisher gauge how crowded a category is before deciding where to launch a new app.

3. Total Installs by Category (Bar Chart)

Chart type: Clustered Bar Chart, Top 10 by installs

Why this visual: A ranked bar chart makes it easy to compare install volume across categories at a glance, ordered by scale.

Key insight: Games is the top category by total installs, followed by Communication and Productivity — reach and app-count leadership do not always match (Games leads both, but the runner-up categories differ from the Treemap's runner-ups).

Business relevance: Shows publishers where user reach is concentrated, which is a different question from where competition is concentrated.

4. Average Rating by Category (Bar Chart)

Chart type: Clustered Bar Chart, Top 15, custom axis range

Why this visual: Ratings across categories cluster tightly (4.0-4.5), so the axis range was narrowed and data labels were added to make small but real differences visible rather than letting them get lost on a 0-5 scale.

Key insight: Category average ratings are closely bunched, with no category dramatically underperforming the rest — quality is fairly consistent across the store.

Business relevance: Tells a publisher that rating alone is not a strong differentiator between categories — other factors (installs, sentiment) matter more for spotting opportunity.

5. Price vs Rating — Paid Apps (Scatter Plot)

Chart type: Scatter Plot, filtered to Type = Paid, X-axis capped at 40

Why this visual: A scatter plot is the clearest way to test whether two continuous variables (price and rating) are related. The chart is filtered to paid apps only, since free apps (about 92% of the dataset) would otherwise dominate the chart as a single cluster at Price = 0, hiding the actual relationship being tested.

Key insight: There is no strong, clear relationship between an app's price and its rating — expensive paid apps are not reliably rated higher than cheap ones.

Business relevance: Suggests that pricing strategy should not be based on an assumption that charging more signals or produces higher perceived quality.

6. Overall Review Sentiment (Donut Chart)

Chart type: Donut Chart

Why this visual: A donut chart summarizes the full sentiment split in one simple visual, before breaking it down further by category.

Key insight: 64% of reviews are Positive, 22% Negative, and 14% Neutral.

Business relevance: Gives a single sentiment health-check number for the whole store, useful as a benchmark when judging any individual category's sentiment mix.

7. Sentiment Distribution by Category (Stacked Bar)

Chart type: Stacked Bar Chart

Why this visual: A stacked bar breaks the overall sentiment split out by category, revealing categories that may look fine on average rating but carry a disproportionate share of negative reviews.

Key insight: Some high-volume categories carry a visibly larger share of negative reviews than others, despite showing broadly similar average star ratings — sentiment analysis surfaces dissatisfaction that a simple rating average can hide.

Business relevance: Flags specific categories where user complaints deserve closer investigation, even if their star rating alone looks acceptable.

7. Interactivity

All three slicers (Category, Type, Content Rating) filter every visual on the page, with one intentional exception: the Price vs Rating scatter plot carries a fixed local filter to Paid apps only, since the free-vs-paid price comparison is the entire purpose of that specific chart.

8. Screenshots

8.1 Dashboard — Default View (Edit Mode)

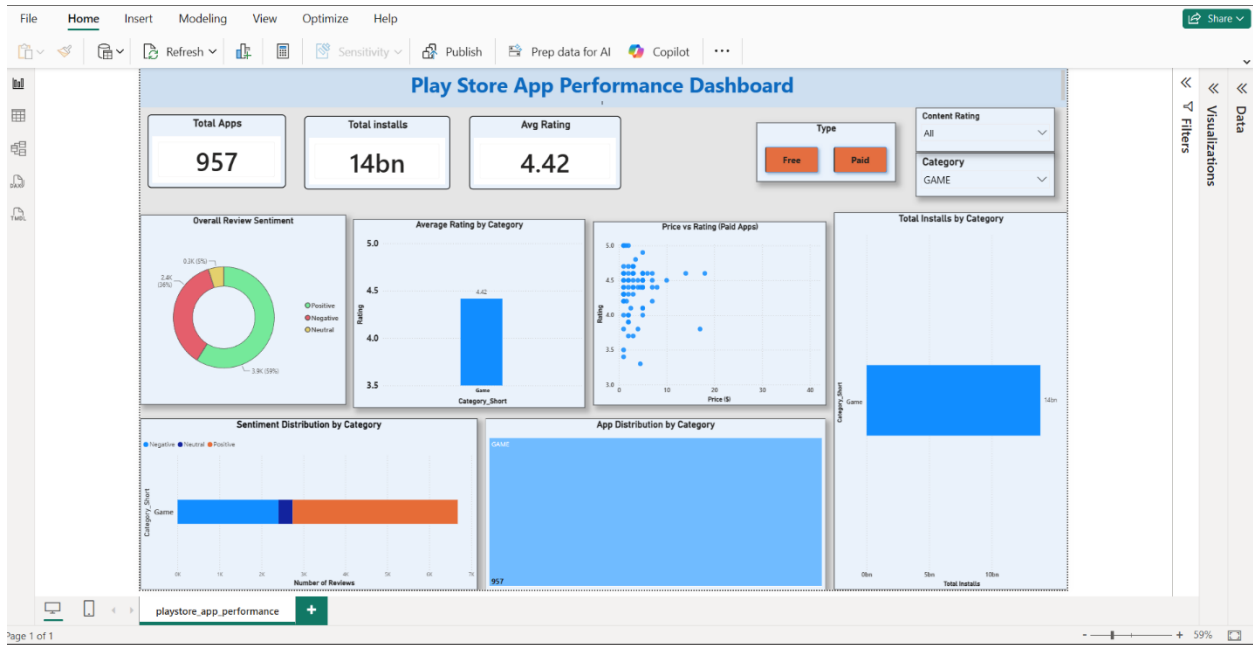


Fig 1: Full dashboard with all slicers set to "All".

8.2 Dashboard — Filtered by Category

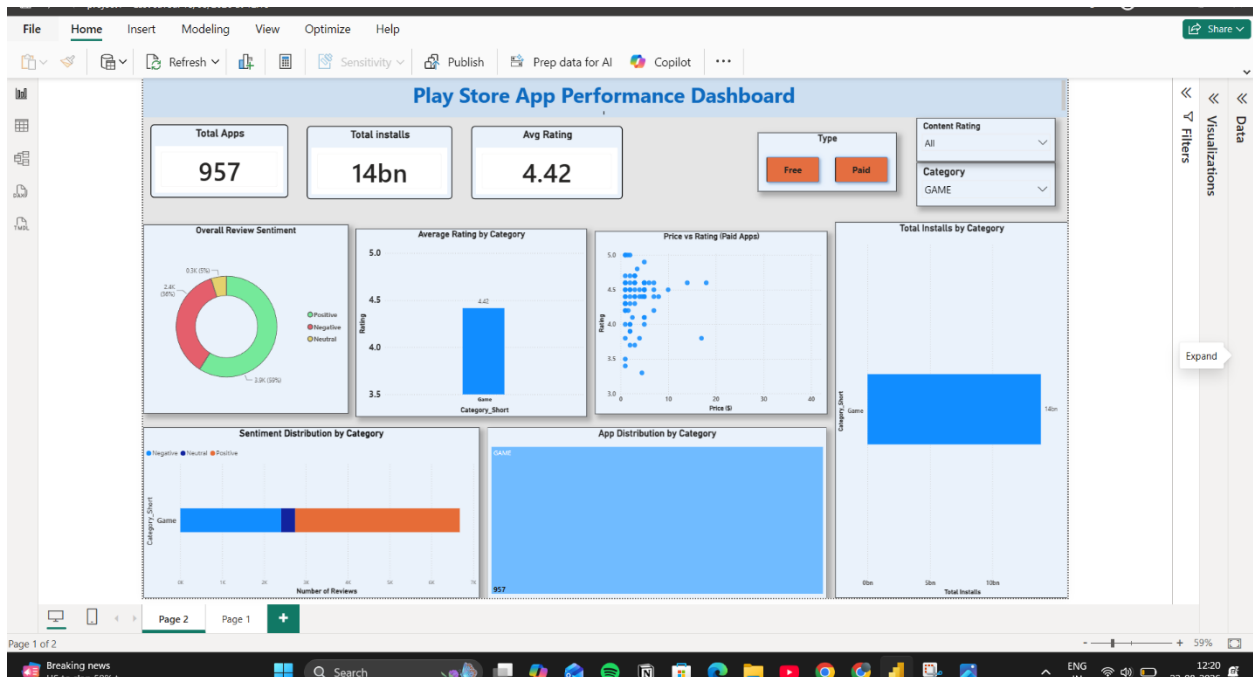


Fig 2: Dashboard filtered to a single category, showing all visuals responding to the Category slicer.

8.3 Dashboard — Presentation View

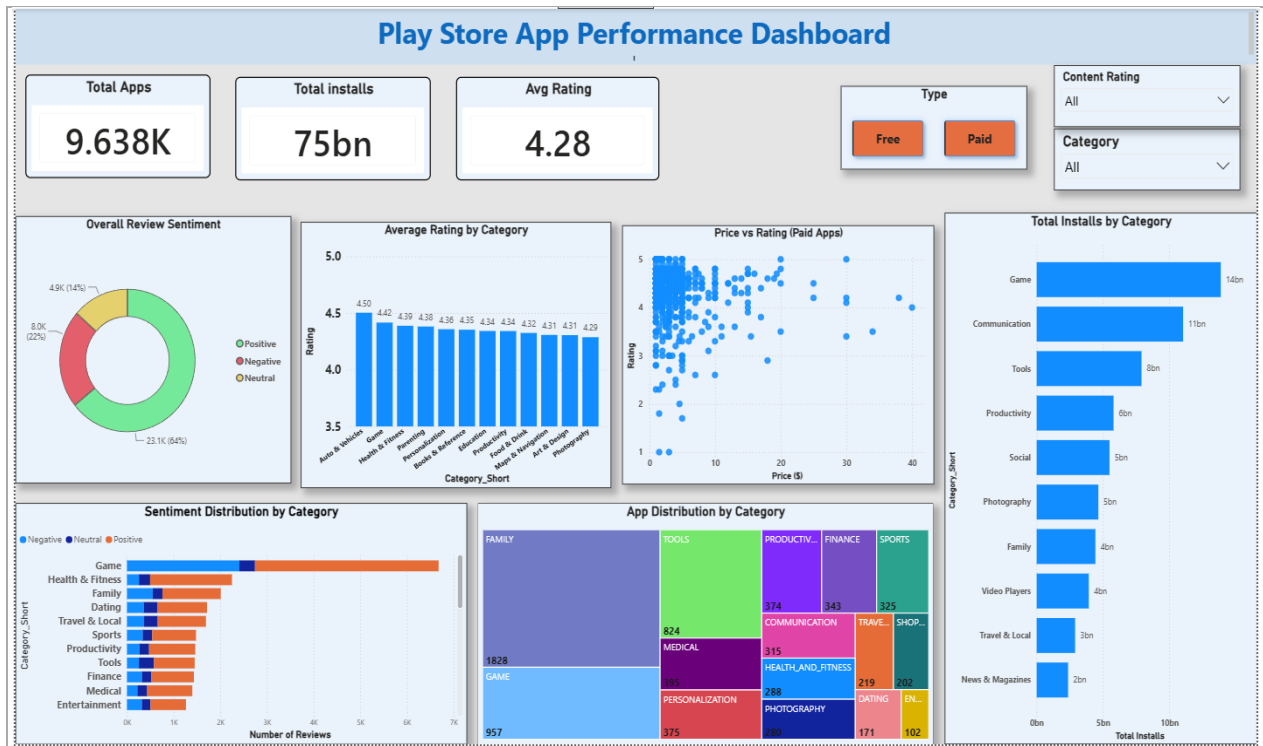


Fig 3: Clean, chrome-free dashboard view, as seen by an end viewer.

8.4 Close-up — Price vs Rating (Scatter Plot)

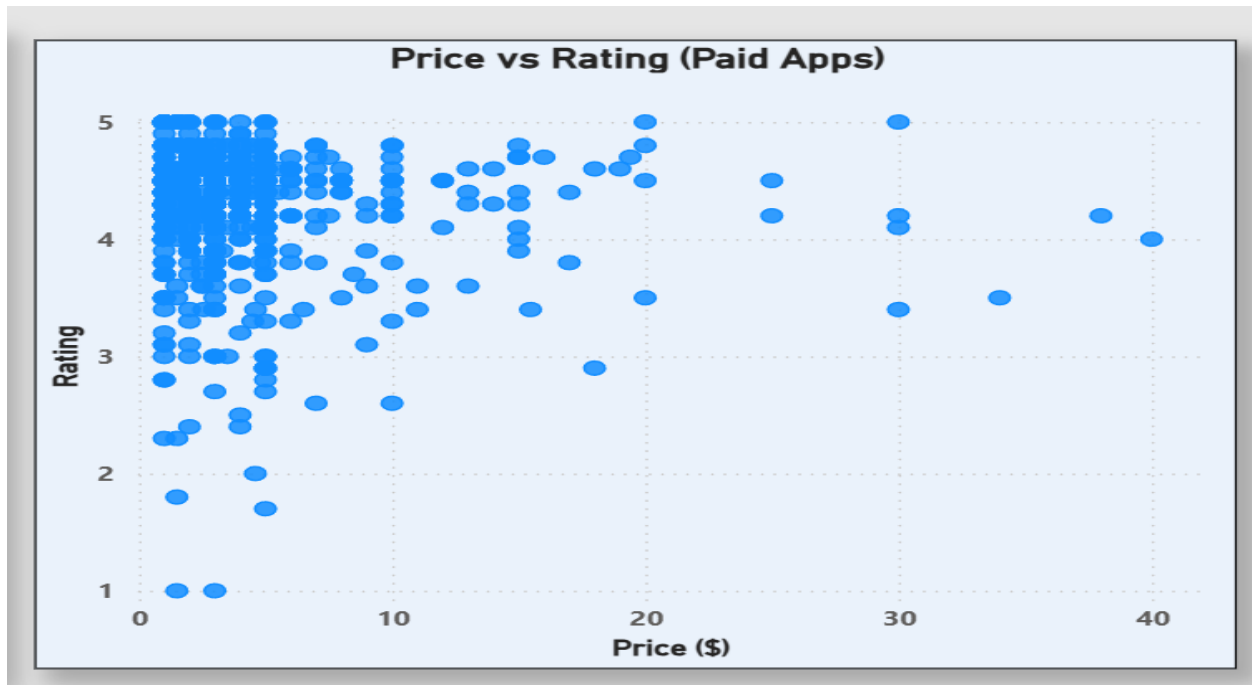


Fig 4: Price vs Rating for paid apps, showing no strong price-quality relationship.

8.5 Close-up — Sentiment Overview

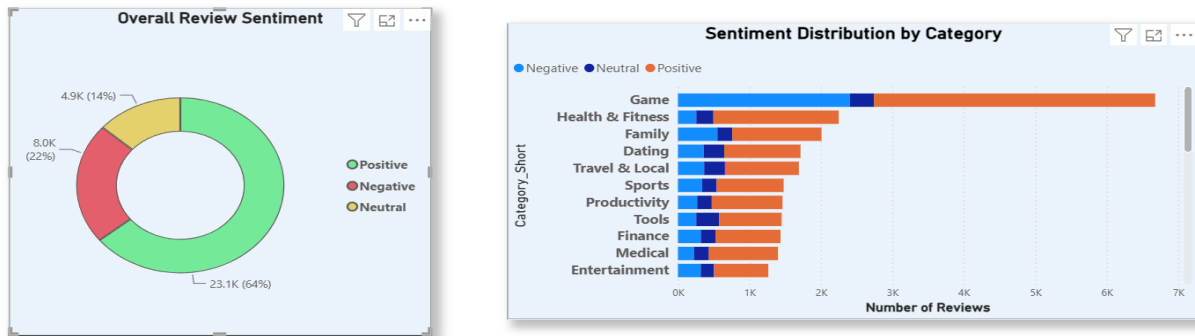


Fig 5: Overall sentiment split alongside its category-level breakdown.

9. Links

- Power BI file (.pbix): included in the project GitHub repository.
- Static PDF export: included in the project GitHub repository.
- GitHub repository: <https://github.com/Dhanush086/playstore-powerbi-dashboard>

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Note: A live Power BI Service shareable link is not included. Power BI Service requires a work/school email domain for account creation, which is not available to me as an independent learner. The .pbix file and PDF export provide full access to the dashboard's design, data model, and DAX measures as an alternative.